

Agentic Self-Healing Virtual Try-On: Overcoming VTO Adoption Barriers through Autonomous Evaluation and Correction

Authors : Bhushan Garware Darshan Barapatre Anibha Athalye

Abstract

Virtual Try-On (VTO) technology holds immense potential for e-commerce by enhancing customer experiences and reducing return rates. However, widespread commercial adoption has been hindered by a critical quality assurance bottleneck: generative AI models frequently introduce semantic errors—such as altering a user's pose or modifying delicate garment attributes—that traditional pixel-level metrics fail to detect. While human evaluation is accurate, it is not scalable. This paper introduces an autonomous Agentic VTO framework. By combining multimodal evaluation with a smart "Self-Heal" loop, the system dynamically generates corrective prompts and routes fixes to specialized models without human intervention. This capability guarantees a monotonic non-degrading pose and outputs an optimized "Champion" image, effectively solving the scalability and trust issues that block VTO adoption.

1. Introduction

With the rapid expansion of e-commerce, Virtual Try-On (VTO) has become a highly sought-after feature aimed at increasing conversion rates and lowering product returns. Advanced VTO systems utilize generative models to synthesize an image of a person wearing a specific product. Despite these advancements, evaluating the fidelity and quality of the generated outputs remains a significant bottleneck.

The evaluation process has historically relied on two flawed paradigms: slow, unscalable manual human evaluation, or fast distribution-level metrics (like FID and SSIM) that lack semantic granularity. A generated image might score well on traditional metrics but still alter the user's pose or change the color of their shoes. To solve this we propose **Agentic Virtual Try-On Engine**, a multimodal AI-based framework for automated evaluation. We extend this concept by introducing a comprehensive tool that not only evaluates but actively self-heals generation errors to produce a production-ready output.

2. The VTO Adoption Problem

For major fashion brands and e-commerce platforms, brand safety and visual fidelity are paramount. The adoption of generative VTO has been stalled by the following challenges:

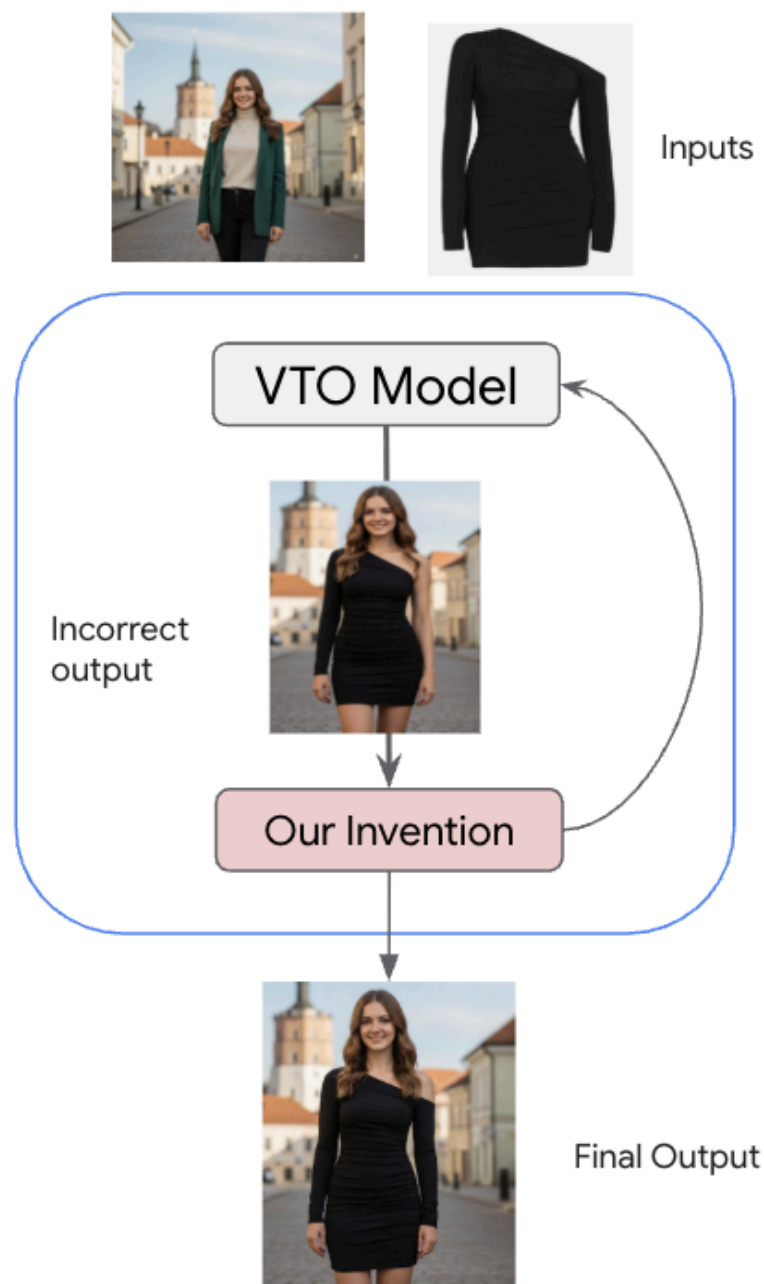
- **Semantic Hallucinations:** Generative models can easily alter the color, omit target garment buttons, or change a user's background.
- **Pose Distortion:** The physical orientation and posture of the subject are often distorted during the VTO process.

- **Unscalable QA:** E-commerce catalogs contain millions of SKUs. Relying on human reviewers to catch subtle hallucinated details (like a missing zipper) is economically unfeasible, yet failing to catch them damages consumer trust.

An automated evaluation framework identifies these errors, but merely flagging a failed generation does not give the user a usable image. To solve the adoption problem, the system must be capable of autonomous self-correction.

3. Methodology: Agentic Self-Healing Framework

The proposed framework integrates Google's [virtual-try-on-001](#) generation model with an autonomous evaluation and recovery engine.



3.1. Automated Multimodal Evaluation

At each generation step, the system performs a dual-pronged evaluation:

1. **Pose Similarity Computation:** Utilizing MediaPipe's `pose_landmarker_heavy`, the system calculates a translation- and scale-invariant pose similarity score between the original person and the VTO output.
2. **Garment and Person Integrity Analysis:** The system uses `gemini-2.5-flash` to inspect specific attributes and detect visual discrepancies between the target garment and the VTO output. It strictly enforces color fidelity, side mirroring, and orientation rules, while actively ignoring accessories like handbags or jewelry during the evaluation.

3.2. Smart Priority Routing

The Agentic VTO loop operates as a state machine that evaluates the generated output against two rigid constraints at each iteration: the pose similarity score (S) and the array of semantic errors parsed from the LLM's JSON output. Based on these evaluations, the system executes one of two mutually exclusive routing paths.

Path A: Structural Pose Repair (`virtual-try-on-001`) If the pose similarity score falls at or below the acceptable threshold 0.90, the system identifies a structural collapse in the generation.

- **Routing Decision:** The task is immediately routed back to the foundational generation model, `virtual-try-on-001`.
- **Prompt Generation:** The Effective Recovery Prompt Engine extracts the specific discrepancies from the previous evaluation and synthesizes a highly constrained, 10-to-15-word corrective prompt. Rather than using static hardcoding, the prompt directly addresses the structural failure identified by the multimodal evaluator (e.g., specific limb placement or occlusion errors).

Path B: Localized Semantic Repair (`gemini-2.5-flash-image` / Nano Banana) If the pose integrity is successfully maintained ($S > 0.90$), but the multimodal evaluator detects garment discrepancies (the attribute differences array is not empty), the system initiates localized semantic repair.

- **Routing Decision:** To correct garment details without destabilizing the successfully generated pose, the system bypasses the primary generator and routes the task to the image-to-image editor, `gemini-2.5-flash-image` (Nano Banana).
- **Input State Constraint:** To force the model to focus strictly on the garment, the agent modifies the context window. It removes the original Person Image entirely and passes exactly two inputs to the model: the Target Garment Image and the *Current VTO Generated Image*.
- **Prompt Generation:** The prompt engine generates localized, concrete visual edit instructions referencing the two provided images. This allows the model to correct specific textures, colors, or missing architectural features (like zippers or collars) directly on the VTO output without altering the base geometry of the subject.

3.3. Dynamic Recovery Prompting and Monotonic Guarantees

Rather than relying on hardcoded phrases, the Effective Recovery Prompt Engine dynamically learns corrective prompts directly from the specific garment attribute differences flagged in the previous iteration.

- For `virtual-try-on-001`, the engine creates concise 10-to-15-word prompts incorporating the learned hints.
- For `gemini-2.5-flash-image`, the engine generates localized visual edit instructions.

To ensure the self-healing process does not introduce new errors, the system enforces a **Monotonic Non-Degrading Pose Guarantee**. It tracks the pose score across all iterations. If a new iteration degrades the pose, the system reverts to the best non-degraded candidate. Ultimately, the system automatically selects the "Overall Champion Output"—the image with the highest pose score and lowest attribute differences across all completed iterations.

3.4. The Monotonic Non-Degrading Pose Guarantee and State Memory

A major risk in iterative image-to-image refinement is that fixing a semantic detail (e.g., repairing a collar) might inadvertently alter the subject's posture, leading to a degraded final image. To prevent this, TryOn-Eval implements a **Monotonic Non-Degrading Pose Guarantee** driven by active state memory.

This mechanism relies on the continuous calculation of the pose similarity score, S , which utilizes translation- and scale-normalized MediaPipe keypoints mapped through an exponential decay function. The state machine tracks this score S across every iteration of the Agentic Virtual Try-On loop.

The guarantee is enforced through two primary mechanisms:

1. **Prompt Clause Retention:** If a generated corrective prompt results in an image that maintains or improves the pose score S , the system flags that specific text clause as "successful". This successful pose clause is retained in memory and automatically prepended to the prompt of the subsequent iteration, ensuring that the model remembers how it successfully stabilized the geometry while it attempts new corrective measures.
2. **State Reversion (Rollback):** If an iteration yields a degraded pose score compared to the previous step, the system immediately rejects the output. The input base is reverted to the best non-degraded candidate from the state history.

By enforcing this strict rollback rule, the system mathematically guarantees that the pose score will never degrade as the loop progresses. Once all iterations are complete—or the target goal of zero attribute differences is reached—the system selects the **Overall Champion Output**. This champion is pulled from the iteration history that achieved the maximum pose similarity score alongside the minimum recorded garment attribute discrepancies.

3.5. Dynamic Learning via the Effective Recovery Prompt Engine

A core innovation of the Agentic VTO framework is its departure from static, rule-based error correction. Instead of relying on a predefined library of hardcoded phrases, the system employs an **Effective Recovery Prompt Engine** that dynamically synthesizes corrective instructions.

The engine achieves this autonomous learning through the following mechanisms:

- **Data-Driven Feature Extraction:** The multimodal evaluation module outputs a structured JSON array containing specific **attribute** and **difference** descriptions. The prompt engine parses this JSON and directly extracts the exact attribute discrepancies—such as "sleeve length," "neckline," "fabric pattern," or "color shade"—detected in the current iteration.
- **Contextual Prompt Generation:** Rather than mapping a generic error category to a static fix, the engine formulates a prompt directly incorporating the descriptive hint learned from the **difference** field. This creates a highly targeted instruction based solely on the unique failure of that specific generation.
- **Iteration Adaptation Strategy:** To overcome stubborn or persistent generation failures, the engine employs an adaptation strategy as the loop progresses. For subsequent attempts (i.e., iteration 2, the engine automatically shifts to significantly different phrasing and applies heightened structural constraints to force model compliance.
- **Architecture-Tailored Synthesis:** The generated prompts are structurally optimized for their destination model's underlying architecture:
 - When routing to the foundational **virtual-try-on-001** model, the engine distills the correction into a concise, 10-to-15-word prompt to efficiently guide the recontextualization without overwhelming the text encoder.
 - When routing to the **gemini-2.5-flash-image** editor, the engine generates localized, concrete visual diff edits that explicitly reference the dual-image input structure.

4. Discussion: Solving the Adoption Barrier

The Agentic Self-Heal mechanism directly neutralizes the primary roadblocks to commercial VTO adoption. By replacing the manual QA process with an autonomous AI agent, brands can scale their VTO catalogs exponentially without expanding human moderation teams.

Furthermore, the Monotonic Non-Degrading Pose Guarantee ensures that edge-case failures are caught and systematically corrected before ever reaching the end-user. By guaranteeing high fidelity in both posture and fine garment details, the self-healing loop delivers the consistency and reliability required for enterprise-grade e-commerce deployments.

5. Conclusion

While modern diffusion models can generate visually stunning virtual try-on images, their tendency to introduce subtle semantic and structural errors has bottlenecked commercial adoption. By uniting the TryOn-Eval diagnostic framework with an Agentic Self-Healing loop, we present an end-to-end pipeline that not only identifies errors but autonomously corrects them using smart model routing and dynamic prompting. This ensures enterprise-ready reliability, removing the final barrier to scalable VTO integration in e-commerce.